Introduction to Optimal Control Problem

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Flight Planning

Different problems

Flight planning problem

- Which path to take between departure and arrival airport?
- Which flight level to choose?
- Which airspeed schedule to use?
- ...

Goals:

- Minimize cost of fuel
- Minimize cost of time
- Minimize overflight fees
- Avoiding adverse meteorology
- Mitigate climate impact
- ...

Flight planning problem - Optimal Control

Flight planning problem

The flight planning problem can be regarded as a

trajectory optimization problem

The most natural way to address a trajectory optimization problem is using

optimal control techniques

Advantages of using optimal control:

- (Non-linear) aircraft dynamics,
- Optimal control law, and
- 4D trajectory.

Change in the current ATM paradigms

TBO (Trajectory-Based Operations) is an ATM approach where flights are managed based on planned, precise flight trajectories. Instead of reactive, real-time adjustments to flight paths, TBO uses **pre-planned trajectories** to predict and coordinate aircraft movement. This enables **better efficiency and capacity management** in the airspace, helping reduce delays, optimize flight paths, and improve fuel efficiency.

Change ATM paradigm:

- Increase capacity;
- Increase ATM efficiency;
- Increase safety;
- Reduce environmental impact.

In the future ATM system the trajectory becomes the fundamental element of a new set of operating procedures collectively referred to as

Trajectory-Based Operations (TBO):

- Accurate 4D trajectories,
- Data sharing,
- Minimal trajectory distortion, and
- Increase in the predictability.

TBO will work best based upon

a constrained paradigm, in which constraints will be issued and the airline will decide the most economical way to meet them.

Business Interests $\rightarrow$ MINIMISE OPERATIONAL COST

Introduction to Calculus of Variations:

The Brachistochrone Problem

Brachistochrone problem

Calculus of Variations

Calculus of Variations

is the name given by Lagrange to the new branch of mathematics that arose as a result of the

Brachistochrone problem.

Recommended paper:

Sussmann, Hector J., and Jan C. Willems. "300 years of optimal control: from the brachystochrone to the maximum principle." Control Systems, IEEE 17.3 (1997): 32-44.

Brachistochrone problem

Brachistochrone problem

A bead descends under the influence of gravity along a friction-less wire,

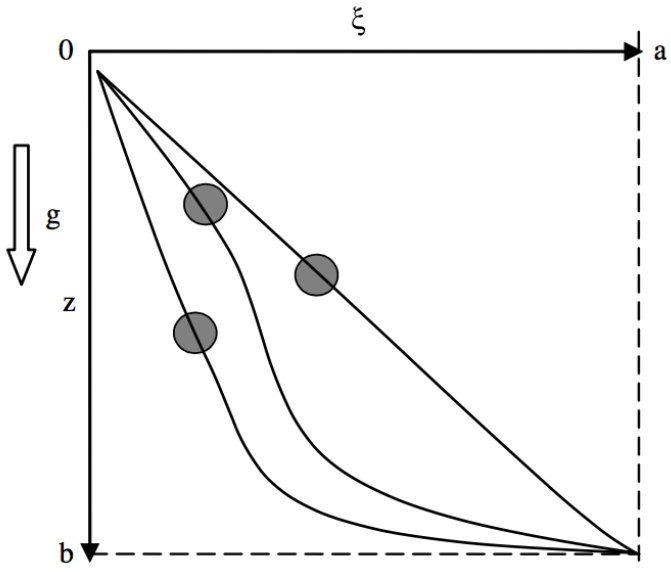


Fig. 3.1 Brachistochrone problem.

the problem is:

to find the shape of the wire for
a minimum time of descent.

https://www.youtube.com/watch?v=Z-qaXZeJT4s

Brachistochrone problem

Brachistochrone problem

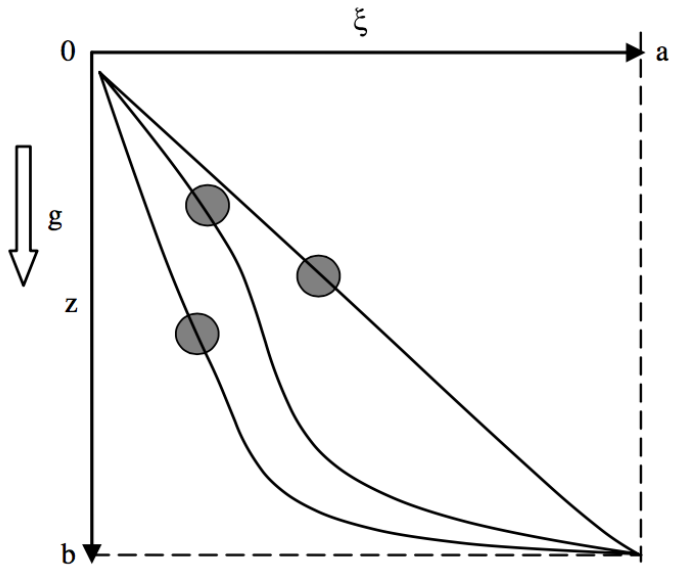


Fig. 3.1 Brachistochrone problem.

If $U(\xi)$ if the velocity of the bead, the kinetic energy of the bead is

$$\frac{1}{2}mU^2(\xi)$$

And, the potential energy of the bead is:

$$mgz(\xi)$$

Brachistochrone problem

Brachistochrone problem

Applying the principle of conservation of energy we have

$$\frac{1}{2}mU^2(\xi) = mgz(\xi).$$

Thus, the expression of the velocity of the bead is

$$U(\xi) = \sqrt{2gz(\xi)}$$

Brachistochrone problem

Brachistochrone problem

If s denotes the distance traveled on the curve, we have

$$U(\xi) = \frac{ds}{dt}$$

Besides,

$$ds^2 = d\xi^2 + dz(\xi)^2 = d\xi^2 + \left(\frac{dz(\xi)}{d\xi}\right)^2 d\xi^2 = \left[1 + z'(\xi)^2\right] d\xi^2$$

Then,

$$dt = \frac{ds}{U(\xi)} = \frac{\sqrt{1 + z'(\xi)^2}}{\sqrt{2gz(\xi)}} d\xi,$$

and the elapsed time to reach the abscissa ξ is

$$t = \int_0^{\xi} \frac{\sqrt{1 + z'(\lambda)^2}}{\sqrt{2gz(\lambda)}} d\lambda.$$

Brachistochrone problem

Brachistochrone problem

The problem is to find the function $z(\xi)$ that minimizes the elapsed time when the bead reaches the point $\{\xi = a, z = b\}$, that is,

$$J(z) = \int_0^a \frac{\sqrt{1 + z'(\lambda)^2}}{\sqrt{2gz(\lambda)}} d\lambda$$

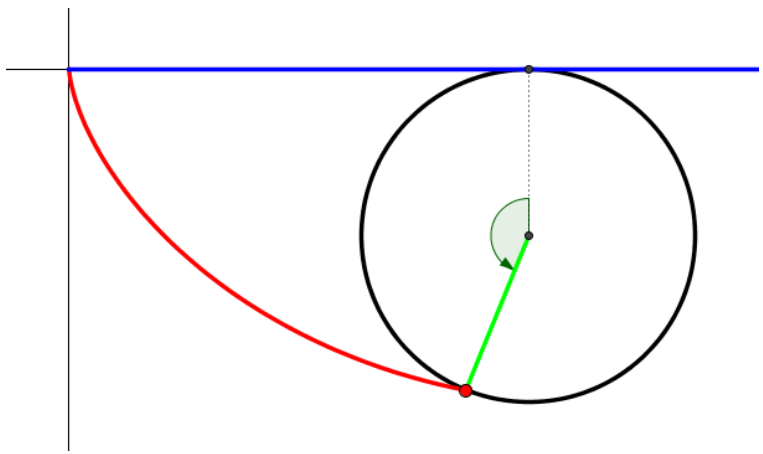
The function is subject to initial and final values, respectively:

$$\begin{aligned} z(0) &= 0, \\ z(a) &= b. \end{aligned}$$

Brachistochrone problem

Brachistochrone problem

The solution is a curve $z(\xi)$ called cycloid, which is also the tautochrone curve (its period is independent of the amplitude of oscillation).



Brachistochrone problem

The simplest problem of the calculus of variations

The generalized version of the brachistochrone problem is euphemistically called

the "simplest problem of the calculus of variations,"

which was considered by Euler and Lagrange. The formulation of the problem is as follows: find a piecewise smooth function
$x : I = [t_0, t_1] \to R$, continuously differentiable with the first derivative continuous, C^1, that minimizes:

$$J(x) = \int_{t_0}^{t_1} f[x(t), x'(t), t]dt$$

subject to initial and final constraints:

$$\begin{aligned} x(t_0) &= x_0, \\ x(t_1) &= x_1. \end{aligned}$$

Introduction to Optimal Control

Optimal Control

Problem (**Simplified Optimal Control Problem**)

$$min\ J(t, x(t), u(t), p) = \int_{t_0}^{t_1} L(x(t), u(t), p)dt;$$

$$s.t.$$

$$\dot{x}(t) = f(x(t), u(t), p),\ \textit{dynamic equations;}$$

$$x(t_0) = x_0,\ \textit{initial boundary conditions;}$$

$$x(t_1) = x_1,\ \textit{terminal boundary conditions;}$$

Optimal Control

Problem (Complete Optimal Control Problem)

$$
\begin{aligned}
&\textit{min } J(t, x(t), u(t), p) = \Phi(t_f, x(t_f)) + \int_{t_0}^{t_f} L(x(t), u(t), p)dt; \\
&\textit{s.t.} \\
&\dot{x}(t) = f(x(t), u(t), p), \ \textit{dynamic equations;} \\
&0 = g(x(t), u(t), p), \ \textit{algebraic equations;} \\
&x(t_0) = x_0, \ \textit{initial boundary conditions;} \\
&\psi(x(t_f)) = 0, \ \textit{terminal boundary conditions;} \\
&\phi_l \leq \phi[x(t), u(t), p] \leq \phi_u, \ \textit{path constraints.}
\end{aligned}
\tag{OCP}
$$

OCP

Objective function

$\min\ J(t, x(t), u(t), p) = \Phi(t_f, x(t_f)) + \int_{t_0}^{t_f} L(x(t), u(t), p)dt;$

$J : [t_0, t_f] \times \mathbb{R}^{n_x} \times \mathbb{R}^{n_u} \times \mathbb{R}^{n_p} \to \mathbb{R}$ is the objective function (Bolza form)

It is expressed as the sum of

- Mayer term $\Phi(t_f, x(t_f))$
- Lagrange term $\int_{t_0}^{t_f} L(x(t), u(t), p)dt$

$\Phi : [t_0, t_f] \times \mathbb{R}^{n_x} \to \mathbb{R}$ and $L : \mathbb{R}^{n_x} \times \mathbb{R}^{n_u} \times \mathbb{R}^{n_p} \to \mathbb{R}$ are assumed to be twice differentiable.

OCP

Variables & Parameters

$t \in [t_0, t_f] \subset \mathbb{R}$ represents time (Notice that the initial time t_0 is fixed and the final time t_f might be fixed or left undetermined)

$p \in \mathbb{R}^{n_p}$ is a vector of parameters

$x(t) : [t_0, t_f] \mapsto \mathbb{R}^{n_x}$ represents the state variables

$u(t) : [t_0, t_f] \mapsto \mathbb{R}^{n_u}$ represents the control functions

$x_0 \in \mathbb{R}^{n_x}$ represents the vector of initial conditions given at the initial time t_0

$\psi : \mathbb{R}^{n_x} \to \mathbb{R}^{n_q}$ provides the terminal conditions at the final time (twice differentiable).

OCP

Variables & Parameters

The system is a DAE system in which the right hand side function of the differential set of equations $f : \mathbb{R}^{n_x} \times \mathbb{R}^{n_u} \times \mathbb{R}^{n_p} \to \mathbb{R}^{n_x}$ is assumed to be piecewise Lipschitz continuous, and the derivative of the algebraic right hand side function $g : \mathbb{R}^{n_x} \times \mathbb{R}^{n_u} \times \mathbb{R}^{n_p} \to \mathbb{R}^{n_z}$ with respect to z is assumed to be regular.
The system must satisfy algebraic path constraints given by the function $\phi : \mathbb{R}^{n_x} \times \mathbb{R}^{n_u} \times \mathbb{R}^{n_p} \to \mathbb{R}^{n_\phi}$ with lower bound $\phi_l \in \mathbb{R}^{n_\phi}$ and upper bound $\phi_u \in \mathbb{R}^{n_\phi}$.
Function ϕ is assumed to be twice differentiable.

But... Why Optimal Control?

Control Theory

Discipline that studies the

behavior of dynamical systems with control inputs

Its aim is to

control the state of the dynamical system in some prescribed manner

Goal of Optimal Control Theory

The goal of optimal control theory is to determine the

control inputs

that will cause a system to

- achieve the control objectives
- satisfying the constraints, and
- optimize some performance criterion

But... Why Optimal Control?

- **Classical control:** Regulate a system around a setpoint (PID, LQR, etc.).
- **Optimal control:** Determine control inputs $u(t)$ that *minimize a cost functional*.

Key question

What is the best possible trajectory and control law, subject to dynamics and constraints?

Applications in Aerospace:

- Minimum-fuel, minimum-time or climate-conscious aircraft trajectories.
- Spacecraft attitude and orbit transfers.
- Entry, descent, and landing optimization.
- Formation flight and path planning.

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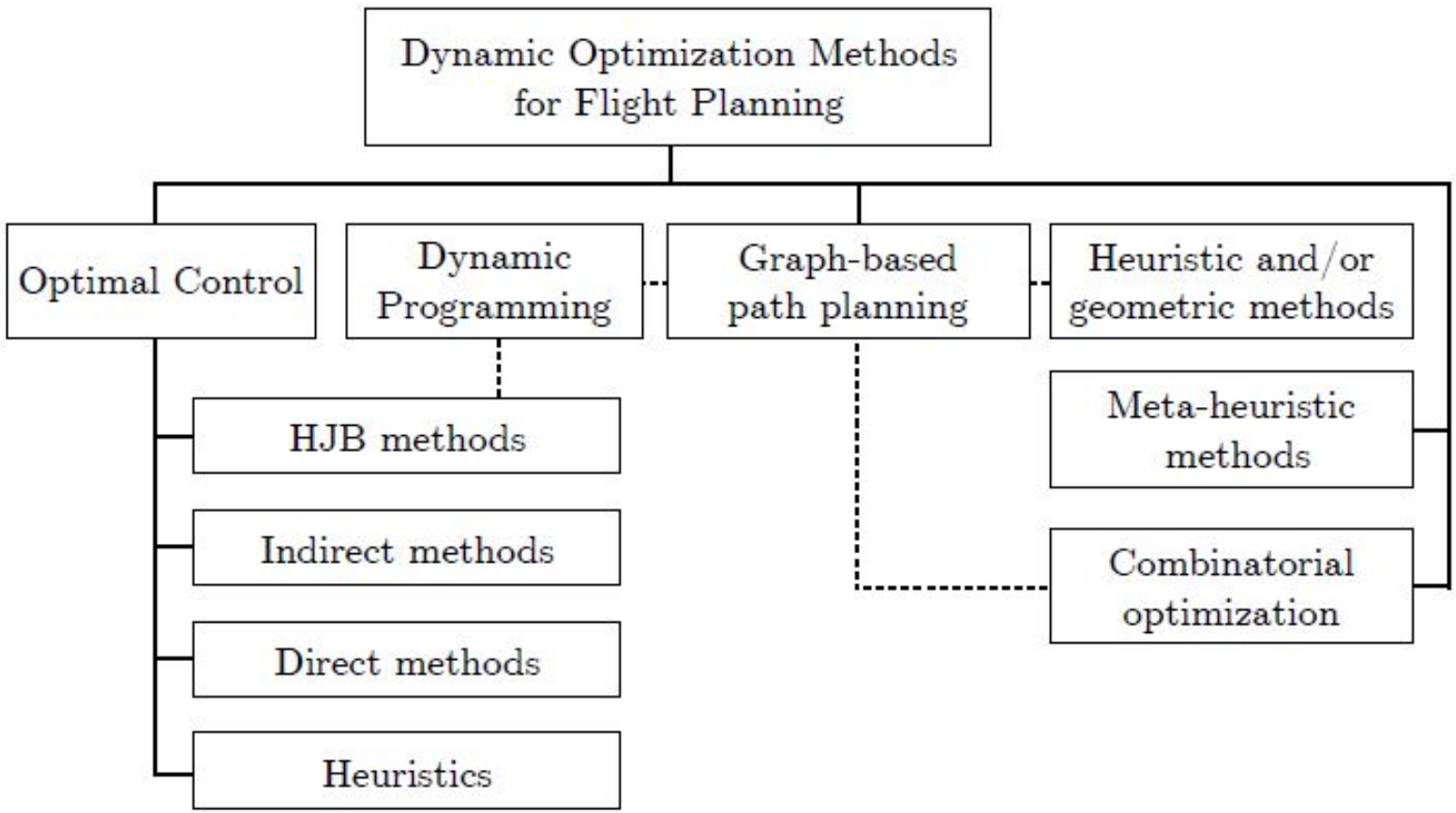
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Optimal Control

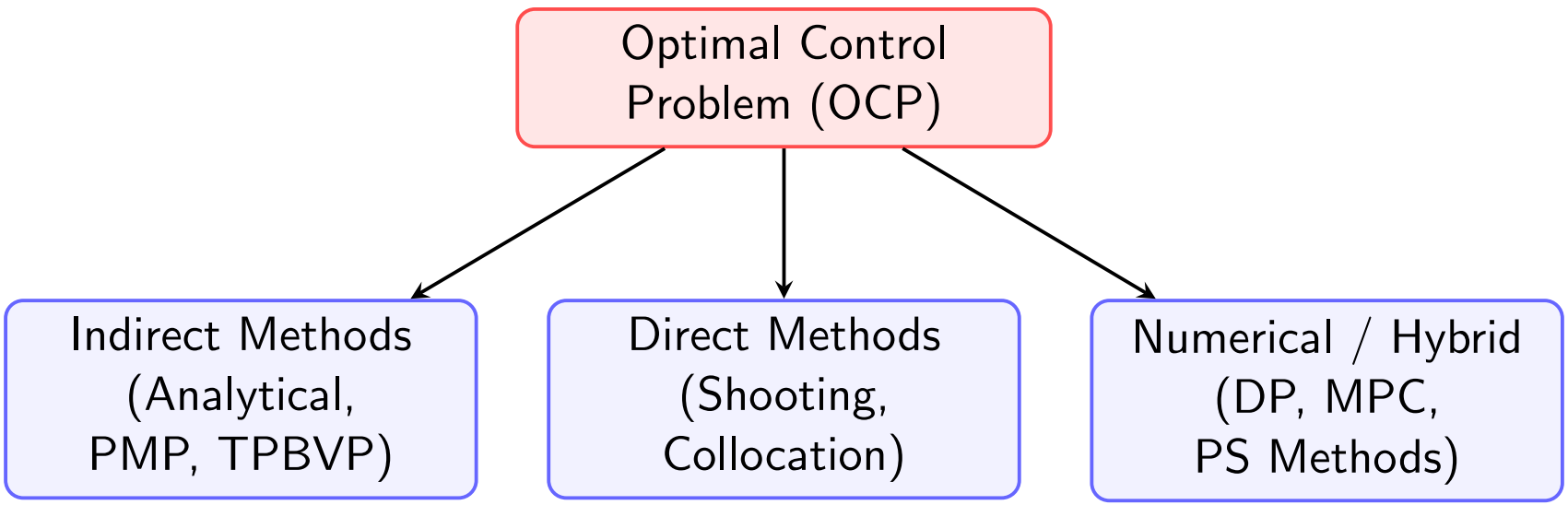
Methodologies

Different problems

Flight planning problem - methodologies



Classification of OCP Solution Methods



Methods for Solving Optimal Control Problems (OCPs)

Overview

There are three main classes of methods to solve Optimal Control Problems:

1. Indirect methods – based on the necessary conditions from optimal control theory.
2. Direct methods – transform the OCP into a nonlinear programming problem (NLP).
3. Hybrid approaches – discretize, approximate, and use computational solvers.

Indirect Methods

Concept

- Derive necessary conditions of optimality from PMP.
- Leads to a two-point boundary value problem (TPBVP).
- Requires analytical expressions of gradients and costates.

Typical Approach

$$\begin{cases} \dot{x} = f(x, u, t), \\ \dot{\lambda} = -\frac{\partial H}{\partial x}, \\ u^*(t) = \arg\min_u H(x, u, \lambda, t), \end{cases}$$

Pros and Cons

- High accuracy, reveals structure.
- Very sensitive to initial guesses and constraints.
- Only for not very complex problems

Direct Methods

Concept

- Discretize states and controls over time grid.
- Replace differential equations with algebraic constraints.
- Solve resulting Nonlinear Programming (NLP) problem using numerical optimization tools.

Two Main Families

- **Direct shooting:** discretize controls, integrate states forward.
- **Direct collocation:** discretize both states and controls simultaneously.

Pros and Cons

- Robust to noise, handles constraints well.
- Less insight into analytical structure.

Numerical and Hybrid Approaches

Examples

- **Dynamic Programming (DP)** – global optimality but exponential complexity.
- **Model Predictive Control (MPC)** – receding horizon online optimization.
- **Hybrid / indirect-direct methods** – use direct methods to initialize indirect ones.

- Out of the scope of this course

Direct Methods:

Pontryagin's Minimum Principle (PMP)

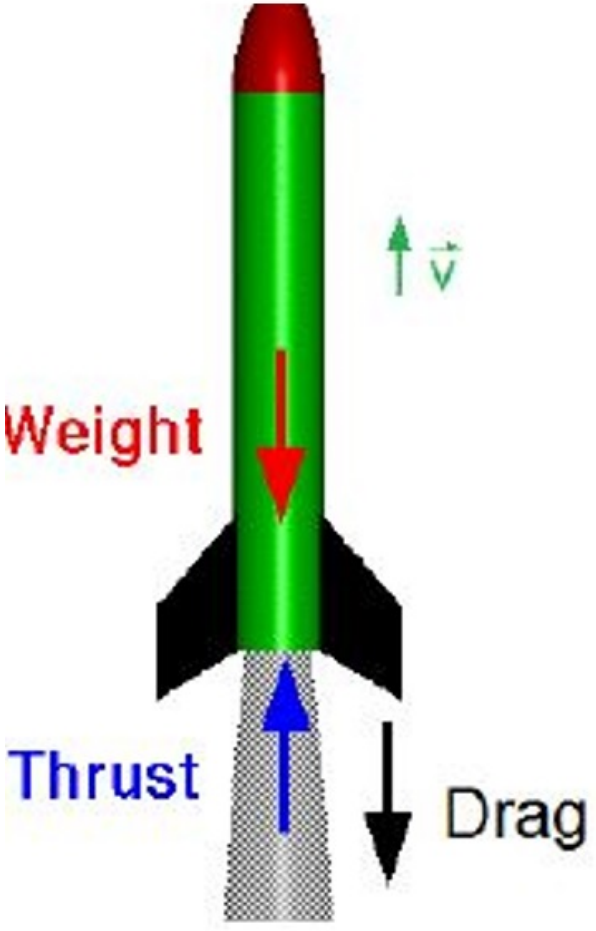
Optimal Control Theory

The first aerospace optimal control problem

Goddard problem

This problem was formulated by R.H. Goddard in 1919:
Maximize the final altitude of a vertically launched rocket, using the thrust as a control and given the initial mass, the fuel mass, and the drag characteristics of the rocket.

Goddard problem equations



Maximize $r(t_f)$

$$\frac{d(mv)}{dt} = T(t) - D(v,r) - mg(r),$$

$$\dot{r}(t) = v(t),$$

$$\dot{m} = -\frac{T}{I_{sp} g_0}$$

$$v(0) = 0, \qquad m(0) = m_0,$$

$$r(0) = r_0, \qquad m(t_f) = m_f$$

$$0 \le F \le F_{max}$$

The equations of motion for the rocket should be normalized.
See, for instance, *Bryson [7, pages 392-394]* for background information.

Goddard problem solution

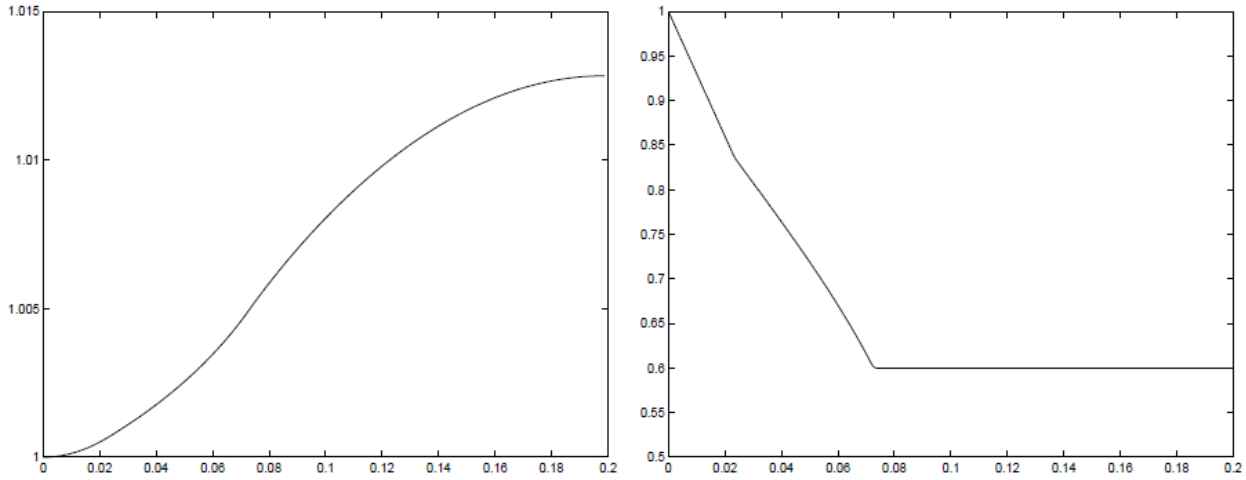


Figure: Altitude and mass for the Goddard rocket problem

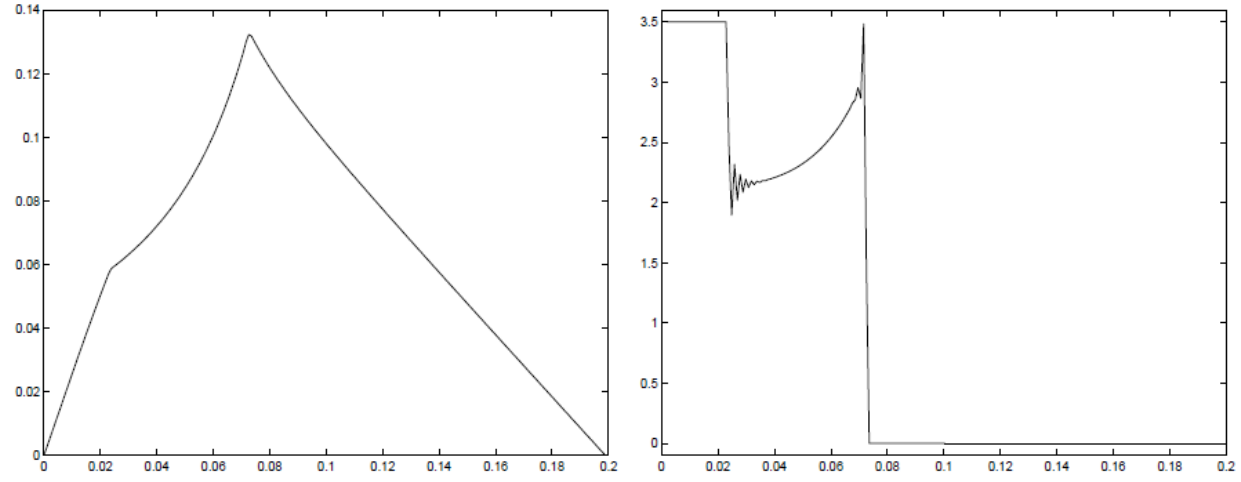


Figure: Velocity and thrust for the Goddard rocket problem

Optimal Control Theory

$$Min \quad J(t, x(t), u(t), p) = \mathcal{M}(t_F, x(t_F)) + \int_{t_I}^{t_F} \mathcal{L}(t, x(t), u(t), p)dt \tag{1}$$

subject to:

$$\dot{x}(t) = f(t, x(t), u(t), p), \quad \text{dynamic eqs} \tag{2}$$

$$0 = h(t, x(t), u(t), p), \quad \text{algebraic eqs} \tag{3}$$

$$x(t_0) = x_I, \quad \text{initial boundary conditions} \tag{4}$$

$$\psi(t_F, x(t_F)) = 0, \quad \text{terminal boundary conditions} \tag{5}$$

$$g_l \leq g(t, x(t), u(t), p) \leq g_u \quad \text{path constraints} \tag{6}$$

See: **Betts, John T. Practical methods for optimal control and estimation using nonlinear programming. Vol. 19. Siam, 2010**.

Optimal Control Theory

Definition (Pontryagin's maximum principle)

Unconstrained problems

A general expression of the necessary conditions of optimality is due to Pontryagin's maximum principle:

$$\mathcal{H}(t, x^*(t), u^*(t), \lambda^*(t)) = \min_{u(t) \in U} \mathcal{H}[t, x(t), u(t), \lambda(t)] \; \forall t \in I = [t_I, t_F]$$

where, the Hamiltonian, $\mathcal{H}$, can be built as follows:

$$\mathcal{H}(t, x(t), u(t), \lambda(t)) = \mathcal{L}(t, x(t), u(t)) + \lambda^T \cdot f(t, x(t), u(t))$$

The PMP says that any optimum of the problem, given by the tuple $(t_f^*, x^*, u^*, \lambda^*)$ will satisfy:

$$u^* = arg \min_u \mathcal{H}[t, x^*(t), u^*(t), \lambda(t)]$$

along the optimal trajectory.

Optimal Control Theory

Unconstrained problems

Definition (Necessary optimality conditions)

$$\frac{d\lambda}{dt} = -\frac{\partial \mathcal{H}}{\partial x}, \quad \text{Euler-Lagrange eqs.,}$$

$$\frac{\partial \mathcal{H}}{\partial u} = 0, \quad \text{the control minimizes the Hamiltonian,}$$

$$\lambda(t_F) = \left[\frac{\partial \mathcal{M}}{\partial x} + \nu \frac{\partial \psi}{\partial x}\right]_{t=t_F}, \quad \text{transversality conditions;}$$

$$\left(\frac{\partial \mathcal{M}}{\partial t} + \mathcal{H}\right)_{t=t_F} = 0, \quad \text{extra transversality condition if } t_F \text{ is free;}$$

State–Adjoint Systems

Motivation

Most numerical optimal control solvers (e.g., *GPOPS-II*, *IPOPT*, ...) internally convert any problem to the **Mayer form**.

- The Mayer form expresses the cost purely as a function of the final state:

$$J = P(x(t_f)).$$

- This allows the entire problem to be represented as a system of **ordinary differential equations (ODEs)** with an **augmented state**.
- Integral costs in Lagrange or Bolza forms are incorporated through an additional state variable:

$$x'_{n+1}(t) = L[x(t), u(t), t], \quad x_{n+1}(t_0) = 0.$$

- The terminal cost then becomes:

$$P(x_f) = x_{n+1}(t_f) + \mathcal{M}[t_f, x(t_f)].$$